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Mind your business... on the cloud

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Recently, we have seen a huge expansion of so-called cloud services, with terms such as SaaS, PaaS, IaaS thrown around seemingly just for the sake of it. For almost any computer need you might have, there now exists an online service that will make your job easier. By letting the cloud do all your messy work, you can now run the entire infrastructure of your business online, and manage it from a simple netbook!

Getting cloudy

As long as you have a device with full internet capability, for most simple purposes you no longer need any other software installed. You can now do your research on Wikipedia, edit and create documents on Google Docs, and email it to whomever you want. This is all old news by now, things have gotten much more interesting.

Cloud services utilise grid computing to distribute computing resources over a wide network (a cloud) and in this process become more scalable and stable. As you are no longer relying on just a single system, chances of downtime are decreased, and system errors are much less catastrophic. By virtualizing the resources and abstracting them, the end-user's experience is much like using any other service, they need not concern themselves with how much resources are coming from where, only that they are available.

Business needs are bound to grow with time as your customer base and services improve, and this is where the cloud is best equipped. Most cloud services allow you to scale your resources at any time, often without needing any downtime. What this means for you is that you no longer need to pay large sums of money for a hosting plan which you may only fully utilize once or twice a year during some special events. Instead you can seamlessly scale your service for the duration of the peak, and thus pay extra only in that duration.

Cloudy services

Two of the forefront services in the cloud arena are the Amazon Web Services, and Rackspace Mosso. They both provide some similar services, but with slightly different approaches.

Amazon web services

Amazon, probably the most famous provider of cloud services such as Amazon EC2 (Elastic Compute Cloud) and S3 (Simple Storage Service) charges users based on how much resources they use instead of having separate pricing blocks.

Simple Storage Service (S3):

The S3 service gives you unlimited online storage capacity, and charges you only for how much data is actually stored on their server, and how much is uploaded or downloaded.

S3 charges you as little as $0.15 (Rs. 7.50) per GB per month of space used, $0.10 (Rs. 5.00) for 1GB data transfer. This means that for storing 10GB of data you will have to pay as little as Rs. 75 a month, with a one time charge of Rs. 50 for when you upload the data the first time. There are additional charges for the number of read write and list requests but they are quite marginal.

It comes with a powerful Access Control System, which allows you to limit access only to specified credentials. You can allow set public read permissions (or even write permissions), allowing you to distribute your files with ease. It can also optionally create torrents of your uploaded files, allowing you to save on distribution costs by letting the users provide some bandwidth.

The data in S3 is organised as buckets. You can create multiple buckets to store your files, and connect to them using your access credentials. The data in these buckets is stored in the form of keys instead of a usual folder and file hierarchy.

Many services have popped up since S3, a popular one being Dropbox.

Elastic Compute Cloud (EC2):

The EC2 service allows you to run a virtual machine on Amazon infrastructure, while giving you fine-grained control of the processing capacity, memory etc., all of which are factored in while calculating the pricing. EC2 charges depend on the operating system and software used by the virtual machine, and the resources allocated to it. So you are paying for exactly the processing and software you use.

There is a plethora of options available to deploy, as Amazon Machine Images (AMI). These are pre-configured (or not) virtual operating system images which can be launched at will. When you launch an EC2 instance, you get SSH terminal access to it, enabling you to remotely configure the system, to install or uninstall any software. It's like renting a computer in some guy's basement all across the other side of the world.

You are charged on a hourly basis based on how much data is received from and sent to the server instance. They even offer virtual machines which come